

Extract from The United Kingdom Merchant Shipping (Accident Reporting and Investigation) Regulations 2012 – Regulation 5:

“The sole objective of the investigation of an accident under the Merchant Shipping (Accident Reporting and Investigation) Regulations 2012 shall be the prevention of future accidents through the ascertainment of its causes and circumstances. It shall not be the purpose of an such investigation to determine liability nor, except so far as is necessary to achieve its objective, to apportion blame.”

#### NOTE

This report is not written with litigation in mind and, pursuant to Regulation 14(14) of the Merchant Shipping (Accident Reporting and Investigation) Regulations 2012, shall be inadmissible in any judicial proceedings whose purpose, or one of whose purposes is to attribute or apportion liability or blame.

© Crown copyright, 2021

You may re-use this document/publication (not including departmental or agency logos) free of charge in any format or medium. You must re-use it accurately and not in a misleading context. The material must be acknowledged as Crown copyright and you must give the title of the source publication. Where we have identified any third party copyright material you will need to obtain permission from the copyright holders concerned.

All reports can be found on our website:

[www.gov.uk/maib](http://www.gov.uk/maib)

For all enquiries:

Email: [maib@dft.gov.uk](mailto:maib@dft.gov.uk)  
Tel: +44 (0)23 8039 5500

## Collision between fishing vessel *Achieve* (HL257) and general cargo ship *Talis*, resulting in the sinking of *Achieve* off Tynemouth, England on 8 November 2020

### SUMMARY

On 8 November 2020, the UK registered fishing vessel *Achieve* collided with the Panama registered general cargo ship *Talis* in fog. The fishing vessel was severely damaged and sank while being towed to port. The cargo ship suffered minor damage. There were no injuries and only minor pollution from *Achieve*.

Image courtesy of [RNLI](https://www.rnli.org/) (Tynemouth)



*Achieve* (HL257)

Image courtesy of Paul Gowen ([shipspotting.com](http://shipspotting.com))



*Talis*

The MAIB investigation identified that neither vessel was keeping an effective lookout in the restricted visibility. Additionally, *Achieve*'s wheelhouse was unmanned at the time of the collision. *Talis*'s radar detected the fishing vessel at close range, but the watchkeeping officer's action was hesitant and too late to avoid a collision.

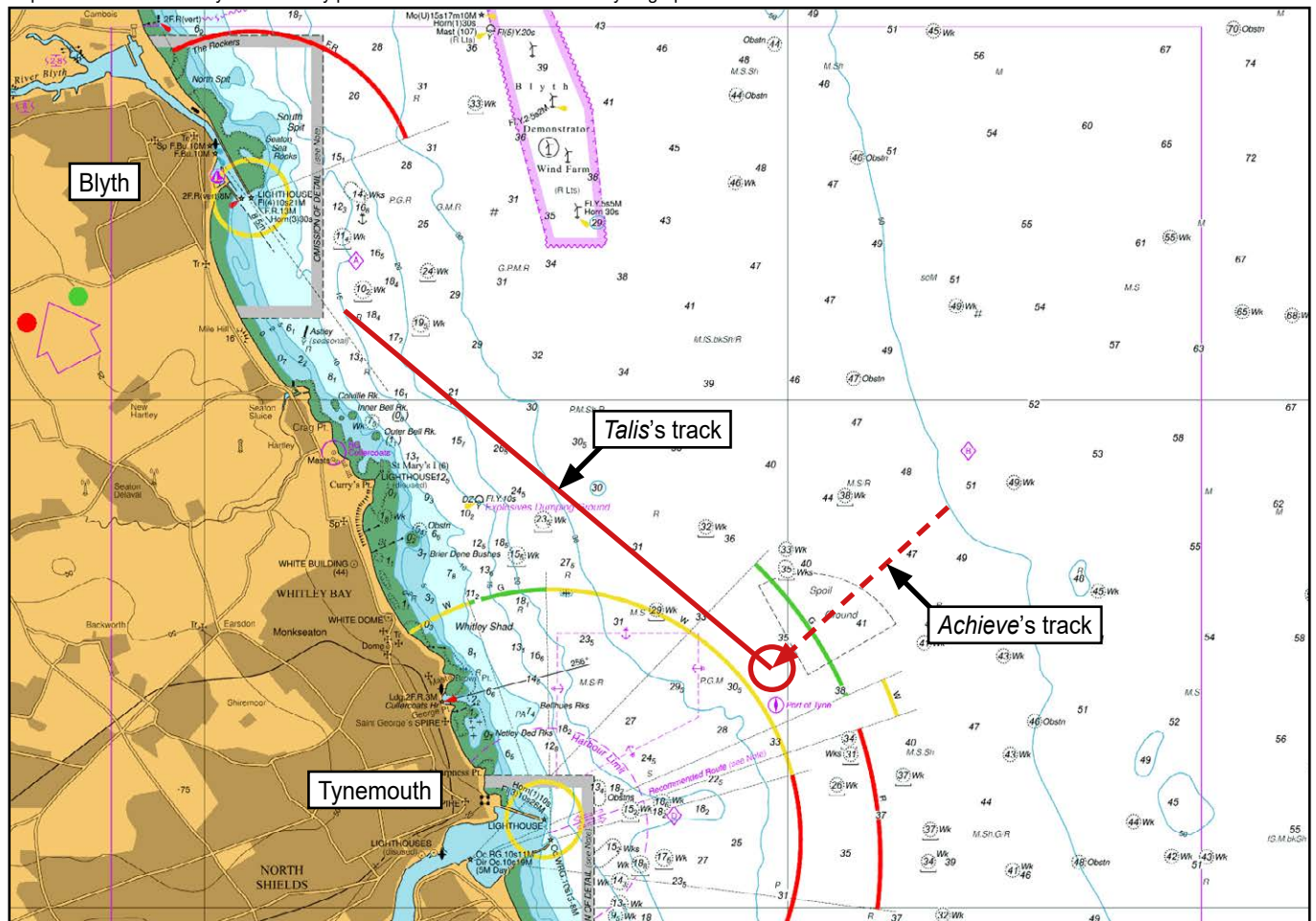
Following the accident, *Achieve* was salvaged and then scrapped. Its owner/skipper, who has since bought a half share in a new boat and fishes part time with its co-owner, has been recommended to put procedures in place to require a proper lookout is kept at all times. A recommendation has been made to *Talis*'s owner to remind its masters and officers of their obligations under the Convention on the International Regulations for Preventing Collisions at Sea 1972, as amended.

### FACTUAL INFORMATION

#### Narrative

At about 0530 UTC<sup>1</sup> on 8 November 2020, the UK registered prawn trawler *Achieve* departed North Shields, England, bound for fishing grounds off Tynemouth (**Figure 1**). A skipper and a deckhand crewed the vessel. The skipper planned to fish during the day and return to North Shields in the afternoon. There was a light southerly wind and it was overcast with fog patches.

<sup>1</sup> Universal time coordinated



**Figure 1:** Extract from Admiralty Chart 156, showing tracks of *Talis* and *Achieve*

The Panama registered general cargo ship *Talis* was in Blyth, England, loading a cargo of red stone for Terneuzen, Netherlands. Cargo operations were completed by 1200 and the ship sailed at 1429. At 1450, the local pilot departed as the ship cleared the main navigation channel. A few minutes later, the master released the helmsman who remained on the bridge as lookout. He set a course on the autopilot to steer 129° and increased the ship's speed to 8 knots (kts).

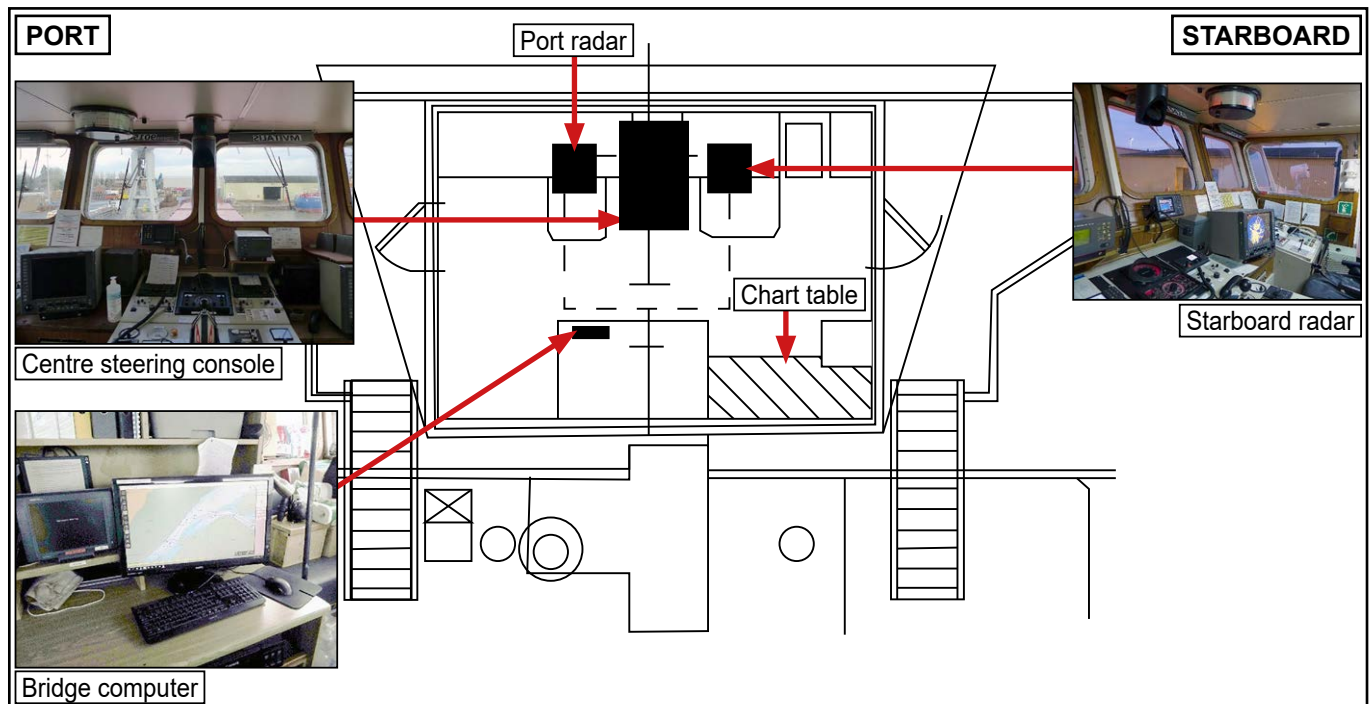
At 1500<sup>2</sup>, *Talis*'s chief officer (C/O) arrived on the bridge to start his watch. The master handed navigation control over to him, then went to the bridge computer to read and send emails. The C/O contacted the duty able-bodied seaman (AB) in his cabin and told him to come to the bridge to keep a lookout. The AB arrived shortly after 1505, and at 1515 the master left the bridge and went to his cabin.

*Talis*'s C/O carried out administrative navigation duties at the chart table and the bridge computer (**Figure 2**). The AB was on the port side between the closed bridge wing door and the centre steering console. The starboard radar was set to the 6 nautical mile (nm) range scale, head-up display, with 1.5-minute trails active. The port radar was in standby mode.

Meanwhile, *Achieve*'s crew had completed their third and final haul of the day, and the skipper set a course for Tynemouth. The vessel's autopilot was engaged, making a course of about 229°, and its speed was about 5kts. During the passage, the skipper familiarised himself with the radar, which was newly fitted two days earlier. As part of this familiarisation he switched between the 1.5nm, 3nm and 6nm range scales. The deckhand was on the aft shelter deck's port side, boxing up the day's catch. As *Achieve* approached Tynemouth, the skipper left the wheelhouse and went to the aft shelter deck to check on the deckhand's progress.

At about 1538, *Talis*'s C/O looked at the starboard radar and observed a target at less than 1nm range, about 30° on the port bow. The C/O reduced the radar range scale to 3nm and checked the automatic

<sup>2</sup> Ship's time was 1600 as it kept Central European Time (UTC+1)



**Figure 2:** Plan of *Talis's* wheelhouse

identification system (AIS) for any signal from the target. Having seen no AIS signal, he instructed the AB to look for a contact and then joined him on the port side by the closed bridge wing door. They both searched visually, the C/O using a pair of binoculars.

Suddenly, the C/O and AB saw *Achieve* emerge from the fog, 30° on the port bow, and, with the aid of binoculars, saw that its wheelhouse was unmanned. He sounded one long blast on the ship's whistle, using the button on the console to the left of the port radar, then went to the centre console and switched the helm to manual control and put the rudder hard-to-starboard.

*Talis's* heading had not appreciably changed when, at 1541, *Achieve* struck the cargo vessel's port side. *Achieve's* skipper and deckhand were thrown to the deck by the force of the collision. The skipper went to the wheelhouse and immediately saw *Talis* disappear into the fog on his starboard side.

### Post collision

On board *Talis*, the C/O immediately put the helm to midships, switched to autopilot, and telephoned the master. The AB watched the fishing vessel drift past the stern until he lost sight of it in the fog. The master arrived on the bridge, pulled the engine control telegraph back to dead slow ahead, and was briefed by the C/O. *Talis's* master followed the guidance in the company emergency response card for a vessel involved in a collision. *Talis* continued on passage at reduced speed while the master made a damage assessment. The hull had suffered minor indentation and handrail damage (**Figure 3**) where *Achieve* had struck the hull. The crew checked the ballast tanks and confirmed to the master that watertight integrity had not been compromised.

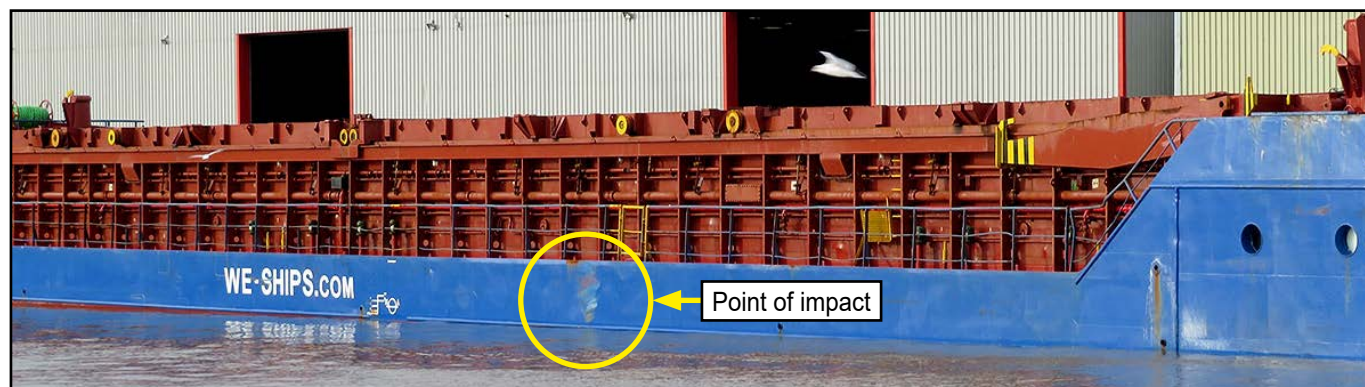
On *Achieve*, the bilge alarms sounded shortly after the collision and both crew checked for water ingress. The deckhand discovered water in the engine compartment at the forward end of the vessel and entered the space to assess the damage. He saw several sprung planks around the stem (**Figure 4**) where water was flowing in. He considered starting the engine-driven bilge pump, but decided not to because its control was at the aft end of the space and he feared being trapped by the flood.

At 1544, *Achieve's* skipper called Humber Coastguard on very high frequency (VHF) radio channel 16 and advised them he had been in a collision and was taking on water. The VHF call was overheard by the *Talis* bridge team. The coastguard immediately tasked the Tynemouth RNLI all-weather lifeboat (ALB) to attend. Meanwhile, the deckhand considered reducing the water ingress by affecting temporary repairs at the stem but assessed it as impractical.



At 1553, the coastguard called *Talis*'s master to ask if he had been in a collision. The master confirmed that he had and, having verified the watertight integrity of his ship, turned the vessel around to return to the collision position and offer assistance to *Achieve*. At 1600, with the lifeboat close to *Achieve*, the coastguard released *Talis* and it resumed its passage towards Terneuzen.

At 1605, the ALB arrived on-scene and transferred a salvage pump to *Achieve*. By 1610, the ALB crew and fishing vessel's skipper had secured a towline to the fishing vessel's bow (**Figure 5**). The ALB then headed to Tynemouth at slow speed with *Achieve* in tow. After about an hour, the fishing boat became unstable as more water flooded the hull and the skipper and deckhand abandoned to the ALB. The lifeboat released the towline and remained close to *Achieve* until, at 1810, it sank, 0.25 miles from the Tynemouth north breakwater in 18m depth of water. The ALB then returned to Tynemouth and landed the skipper and deckhand.



**Figure 3:** Damage to *Talis*'s port side



**Figure 4:** Damage to *Achieve*'s stem



**Figure 5:** *Achieve* being towed by RNLI all-weather lifeboat

Photographs (Figures 4 and 5) courtesy of RNLI (Tynemouth)

## ***Talis***

Built in 1992, *Talis* was an 81.8m long, 1495gt single hold general cargo ship. It was employed on time charter and carried various dry cargoes, unitised and in bulk, around northern Europe.

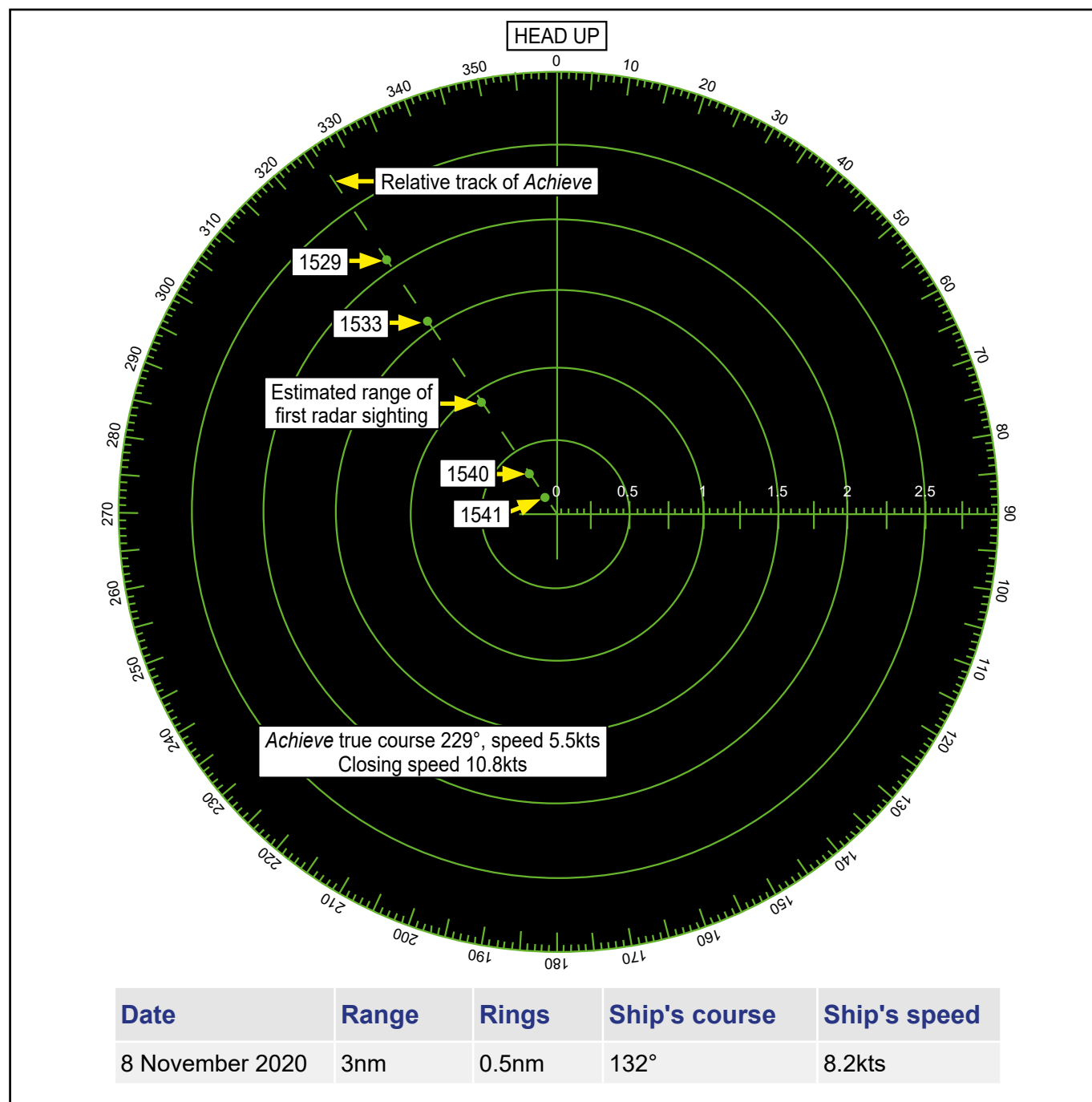
The vessel was managed by WeShips Denizcilik ve Ticaret A.Ş. (WeShips), Turkey. WeShips was founded in February 2020 and, in addition to *Talis*, managed six other ships between 1495gt and 2873gt. The company operated a safety management system (SMS) issued on 22 July 2020 in accordance with the requirements of the International Safety Management (ISM) Code.

*Talis*'s bridge was equipped with two Furuno 1510 Mk3 radars, two global positioning system (GPS) receivers, and an AIS unit. The vessel used paper charts as its primary means of navigation; an electronic chart system (ECS) with GPS input was available through the bridge computer. The ECS was used primarily for passage planning and route overview.

*Achieve*'s skipper had been a fisherman for more than 40 years and had owned *Achieve* for 11 years. The deckhand had worked with the skipper on board the boat for four years. Both the skipper and the deckhand had completed all UK statutory safety courses<sup>4</sup>. In addition, the deckhand held a non-mandatory under 16.5m skipper's certificate<sup>5</sup>.

## Position data

A bulk carrier anchored off Tynemouth, 1nm away from the accident position, was fitted with a voyage data recorder (VDR)<sup>6</sup>, and MAIB recovered its data for analysis. Radar information was recorded and used to determine *Talis* and *Achieve*'s positions before the collision. MAIB plotted the positions (**Figure 6**) to determine the vessels' aspect and *Achieve*'s relative movement, as seen from *Talis*, prior to the collision.



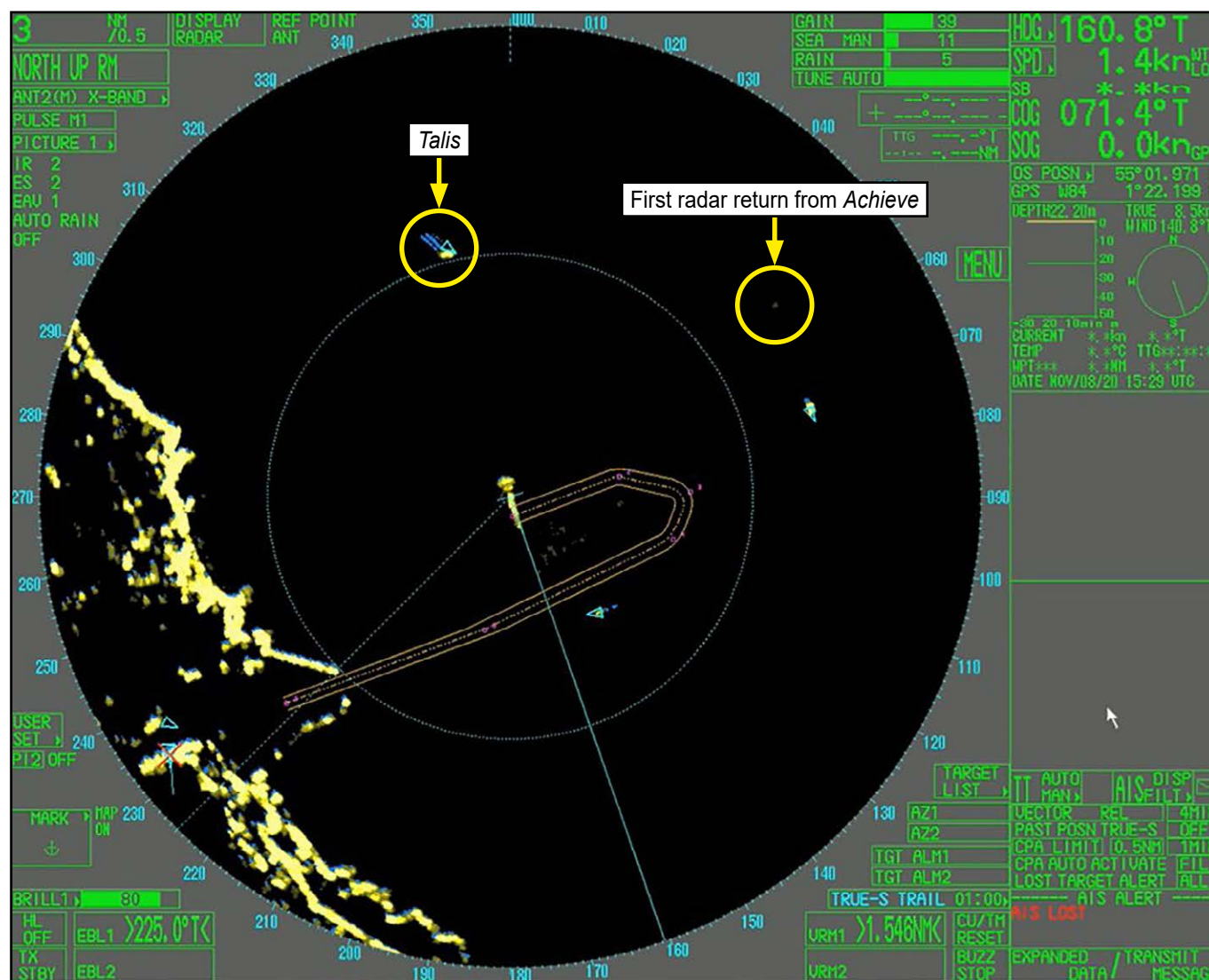
**Figure 6:** Plot of *Achieve*'s positions relative to *Talis*, recreated from third party VDR

<sup>4</sup> Anyone starting work on a UK registered commercial fishing vessel must complete one-day basic safety training courses for sea survival, firefighting and prevention, first aid, and health and safety. Experienced fishermen (those with two years' experience) also need to complete a further one-day course in safety awareness and risk assessment.

<sup>5</sup> MCA approved Seafish Skipper's Certificate for Under 16.5m Vessels up to 20 miles off-shore.

<sup>6</sup> *Talis* was not required to carry a VDR. It is required to be fitted to vessels 3000 gross tonnage and upwards.

A faint radar return appeared on the bulk carrier's radar at a range of 2nm (**Figure 7**) and sporadic returns were subsequently seen. Following the collision, *Achieve's* radar return became much stronger as its aspect to the bulk carrier changed.



**Figure 7:** VDR radar image, showing first radar return of *Achieve* at 1529 UTC

## Regulations and guidance

Both vessels were required to comply with the COLREGs.

Rule 5 (Lookout) stated that:

*Every vessel shall at all times maintain a proper lookout by sight and hearing as well as by all available means appropriate in the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.*

Rule 19 (Conduct of vessels in restricted visibility) applied to vessels not in sight of one another when navigating in or near an area of restricted visibility. Rule 19(d) stated that:

*A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time...*



Furthermore, Rule 35 (Sound signals in restricted visibility) required that both vessels make sound signals while in or near an area of restricted visibility. As defined by the COLREGs, both *Achieve* and *Talis* were 'power-driven vessels' and were required to sound one prolonged blast on the vessel's sound signalling device every two minutes.

Before the collision, neither vessel was making the required sound signals. *Talis*'s C/O did not switch on the automatic fog signal as he assessed that there was little or no traffic in the area. *Achieve*'s skipper had not considered making sound signals.

### **Marine Guidance Note 313 (F) Keeping a safe navigational watch on fishing vessels**

The UK Maritime and Coastguard Agency (MCA) noted that '*Investigations into collisions, groundings and near misses involving fishing vessels had continued to show that poor watchkeeping was a major cause.*' As a result, MGN 313 was published in 2006, which gave guidance to fishermen on keeping a safe navigational watch and included the following factors:

- *The wheelhouse must not be left unattended at any time.*
- *The weather conditions, visibility and time of day.*
- *The use and operational condition of navigational aids such as radar, echo sounder, automatic pilot, and position-fixing equipment(s). [sic]*

### **Marine Guidance Note 369 (M+F) Navigation in Restricted Visibility**

Continued misunderstanding of the COLREGs by navigating officers prompted the MCA to publish Marine Guidance Note 369 (M+F) in 2008, entitled *Navigation in Restricted Visibility*.

The guidance note:

- *Describes the proper conduct of vessels in restricted visibility.*
- *Sets out how to apply the Rules to determine risk of collision in a close-quarters situation and decide on the correct avoiding action.*
- *Reminds operators that Sections I and III of the Steering and Sailing Rules of the Collision Regulations must be complied with strictly.*
- *Advises operators on how they should determine a safe speed and a close-quarters situation in restricted visibility.*

### **International Chamber of Shipping Bridge Procedures Guide**

Annex 3, Section C – Emergencies – of this publication contained Checklist C4, which provided guidance on the steps to be taken in the event of a collision. *Talis*'s SMS contained an emergency response card that closely followed the checklist within the Bridge Procedures Guide.

## **ANALYSIS**

### **The collision**

*Achieve* and *Talis* collided in fog because neither watchkeeper was keeping an effective lookout: the wheelhouse on board *Achieve* had been left unattended, and the C/O onboard *Talis* was distracted from lookout duties. Once *Achieve* was detected on board *Talis*, the decision to alter course was taken too late to avoid the collision.

### **Likely detection ranges**

As the two vessels approached each other, the aspect that *Talis* presented to *Achieve* would have provided a strong radar echo return. With the radar set on the 6nm range scale, *Achieve*'s skipper should have detected *Talis* about 30 minutes before the collision or, if set on the 3nm range scale, about 15

minutes before the collision. However, it could not be determined what effect the fog or control settings had on the radar's performance as this was the first day since its fitting that the radar was being used at sea and the skipper was still familiarising himself with its use.

*Achieve* was detected by the radar on board the bulk carrier at anchor off Tynemouth at a range of 2nm at that time when the fishing vessel's aspect provided for the weakest radar signal echo reception. The aspect that *Achieve* presented to the approaching *Talis* should have provided a better radar return on *Talis*'s radar, so detection at a range greater than 2nm should have been possible. *Achieve* was a wooden fishing vessel, so it is probable that, had the portable radar reflector on board been mounted on the vessel's mast, it would have increased the strength of the returning echo signal to *Talis*, thereby improving the chance of earlier detection.

Neither vessel was making sound signals, which could have alerted them to the other's presence. While the aural detection of a fog signal would not have been an effective method of determining range and bearing, or of the risk of collision, it would have indicated that another vessel was close and that immediate action was required.

## Lookout

Navigation in restricted visibility requires heightened vigilance because visual observation is often negated. The bridge team's role and the proper use of bridge equipment are crucial to provide an early warning of potential dangers with the radar, in particular, becoming the 'eyes' of the watchkeeper. How the radar is operated and monitored in or near areas of restricted visibility is key to maintaining a proper and effective lookout.

Once *Talis*'s C/O had taken the con from the master, he checked the radar and AIS and saw no traffic to concern him. Consequently, his distraction at the computer and chart table, while undertaking administrative duties, meant that he missed the detection of *Achieve*'s radar return until it was less than 1nm away. At that range, and with a closing speed of about 11kts, it gave the C/O about five minutes to assess the risk of collision and take avoiding action.

*Achieve*'s skipper had set his course to Tynemouth, which took the vessel close to the Port of Tyne pilots' boarding area and ship anchorage. During the passage, given the restricted visibility, his frequent use of radar would have been vital in assessing any traffic conflicts.

The skipper's decision to leave the wheelhouse unattended was dangerous given the vessel's location, its direction of travel, and the prevailing restricted visibility. It resulted in him being unable to keep a proper and effective lookout.

## Determining the risk of collision

Following the first detection of *Achieve* on the radar, *Talis*'s C/O sought visual confirmation of the target rather than plotting it and continuing to monitor his radar. He had been relying heavily on AIS to provide him with warning of approaching traffic and, without any correlating AIS transmissions, he was unsure of the radar echo's validity. By looking for the contact visually instead of plotting its relative movement on the radar, the C/O denied himself vital information about the relative track of the fishing vessel and the risk of collision.

*Talis*'s bridge team estimated that *Achieve* was sighted at 0.4nm and about 30° to 40° off *Talis*'s port bow. The combined closing speed of the two vessels was about 11kts and would have allowed less than two minutes to consider the collision risk. In the mistaken belief that the fishing vessel would alter its course, the C/O continued to look at the vessel with binoculars rather than take immediate action to avoid the collision.



## Keeping a safe navigational watch

*Talis*'s SMS was clear on how to conduct watchkeeping and bridge procedures; however, these were not fully implemented by the C/O. In addition to the infrequent use of the radars, and no use of the fog signal required by the COLREGs, he was acting as the sole officer on the bridge. If the SMS requirement to have two officers on the bridge in restricted visibility had been followed, there would have been greater opportunity to use all available means to keep a good lookout. This would have encompassed proper use of the radars, including long-range scanning to obtain early warning of risk of collision, and radar plotting or equivalent systematic observation of detected objects.

On *Achieve*, the MGN 313 (F) principles of keeping a safe navigational watch were not followed; leaving the wheelhouse unattended to check on the progress of the deckhand in processing the catch was particularly unnecessary. The skipper was not keeping a proper lookout on the vessel's radar nor was *Achieve* making the required sound signals for vessels in or near restricted visibility areas.

## Action to avoid collision

The radar echo of *Achieve* that *Talis*'s C/O identified did not prompt him to take early action to either alter course or reduce the vessel's speed. Occasionally, radars produce false echoes, but these can often be identified by altering the vessel's heading. Where there was some degree of uncertainty regarding the validity of the radar echo seen on *Talis*, it would have been prudent to take early action, by either altering course or reducing speed, in accordance with COLREGs Rules 19(d) and 19(e).

By the time *Talis*'s C/O saw that *Achieve*'s wheelhouse was unmanned, both vessels were close to each other and immediate action should have been taken to avoid a collision. Given the C/O's uncertainty of *Achieve*'s intentions, the sounding of one long blast on the ship's whistle was less appropriate than five short and rapid blasts<sup>7</sup>. However, with no-one in the fishing vessel's wheelhouse there was little prospect of a swift reaction and using the whistle control, located on the port side of the bridge, delayed the C/O changing the helm to manual control and putting the rudder to starboard.

## Actions following collision

*Achieve*'s skipper was prompt in calling Humber Coastguard to inform them of the collision and his predicament with flooding.

It is unclear whether *Talis*'s bridge team tried to contact *Achieve*. However, they overheard the call made by *Achieve*'s skipper to Humber Coastguard and, following their checklist, verified the watertight integrity and safety of their ship before, having been called by Humber Coastguard, returning to *Achieve* to offer assistance.

The RNLI ALB's rapid response and attendance allowed positive action to deploy a salvage pump, and to standby in case of worsening conditions.

*Achieve*'s skipper and the ALB coxswain's decision to tow the vessel was understandable given the port's proximity, and towing from the bow was the quickest method. However, this may have increased the water inflow, which resulted in the vessel sinking.

---

<sup>7</sup> COLREGs Rule 34 (d) 'When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle.'

## CONCLUSIONS

- Neither crew was keeping an effective lookout in the prevailing conditions of restricted visibility.
- *Talis* did not follow the company's SMS requirements to have two officers on the bridge during periods of restricted visibility, and the C/O allowed himself to become distracted by administrative work instead of monitoring the radar.
- *Achieve's* skipper was not in the wheelhouse in the lead up to the collision so could not keep a proper and effective lookout.
- Had *Achieve's* radar reflector been mounted on the mast it may have improved its radar echo strength and, therefore, visibility on *Talis's* radar.
- *Talis's* C/O relied heavily on AIS information to provide early warning of approaching traffic, and lost valuable time validating a weak radar contact. *Achieve* took no action to avoid the collision, and the action taken by *Talis's* C/O was too late.
- Neither vessel was sounding fog signals, which may have alerted them to the other's presence.

## ACTION TAKEN

### Actions taken by other organisations

**WeShips Denizcilik ve Ticaret A.Ş.** has:

Reviewed its safety management system regarding bridge manning in restricted visibility and recognised that the requirement was aligned for ships with a master and three watchkeeping officers. The company has amended the SMM to reflect the manning level of master and two watchkeeping officers on board its vessels.

## RECOMMENDATIONS

***Achieve's* owner/skipper** is recommended to:

- 2021/132** Ensure that policies and procedures are put into place on any future vessels he might own or skipper that clearly state the obligation to keep a proper lookout at all times, as required by the COLREGs.

**WeShips Denizcilik ve Ticaret A.Ş.** is recommended to:

- 2021/133** Issue a fleet safety bulletin to remind its masters and navigation officers of their obligations to comply with the COLREGs, particularly the requirements of Rule 5 (Lookout) and Rule 19 (Conduct of vessels in restricted visibility).

Safety recommendations shall in no case create a presumption of blame or liability